Year 9 Science

Plate Tectonics

Plate tectonics is the theory that the earth's crust is divided into several plates (about 30). It is the movement of these plates that caused Pangaea to break apart, and continues to cause the drifting of the continents. But what then causes the tectonic plates to move?

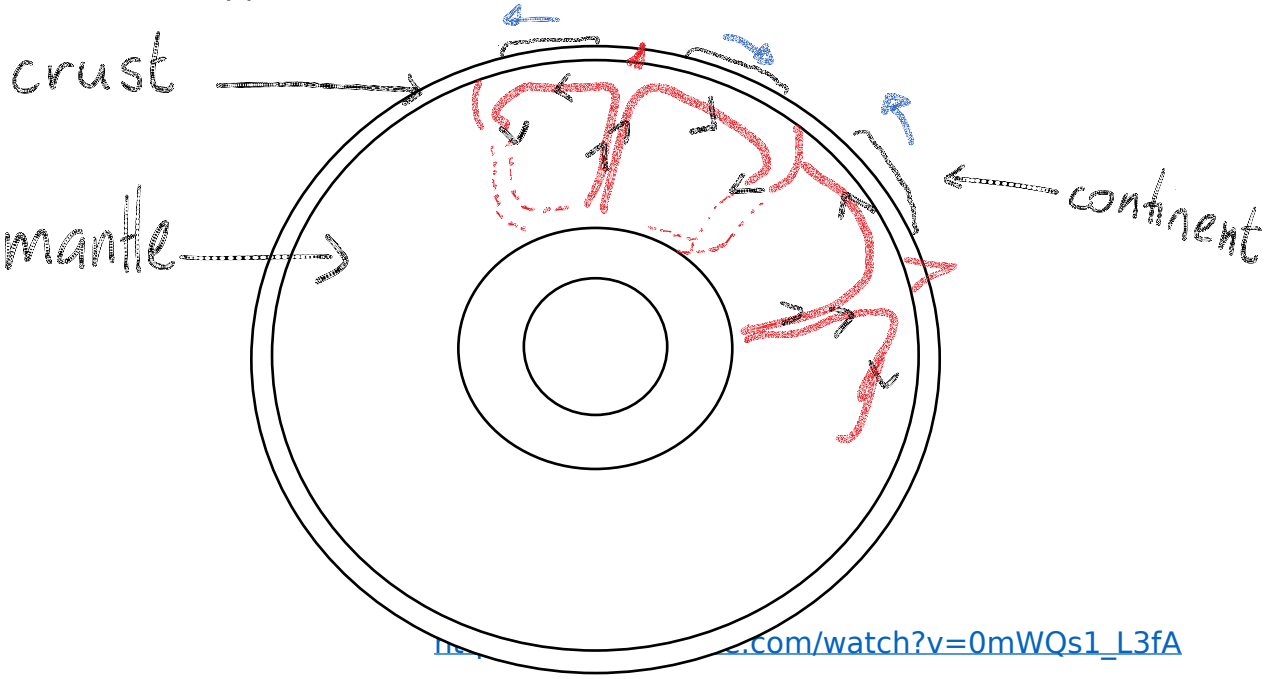
Watch the video carefully and answer the following questions. You also need to use your textbook (Chap 8.2, page 296 – 298) for some of these answers.

Earth's layers

1. Which two elements make up most of the inner core?
   Iron and nickel
2. Which layer is the thickest?
   The mantle
3. The upper mantle and crust make up the *litho*sphere.
4. Describe the appearance of the earth's outer shell.
   Normally covered in water and vegetation. Without these, it appears cracked into tectonic plates. Each plate carries a continent, ocean basin or both.

Movement of tectonic plates

1. Using the diagram, sketch what happens in the **mantle** that causes the movement of the tectonic plates – and therefore the **continents** – of the upper mantle and **crust**.



[...].com/watch?v=0mWQs1_L3fA

2.

a. Describe a divergent boundary.

Plates that are moving away from each other

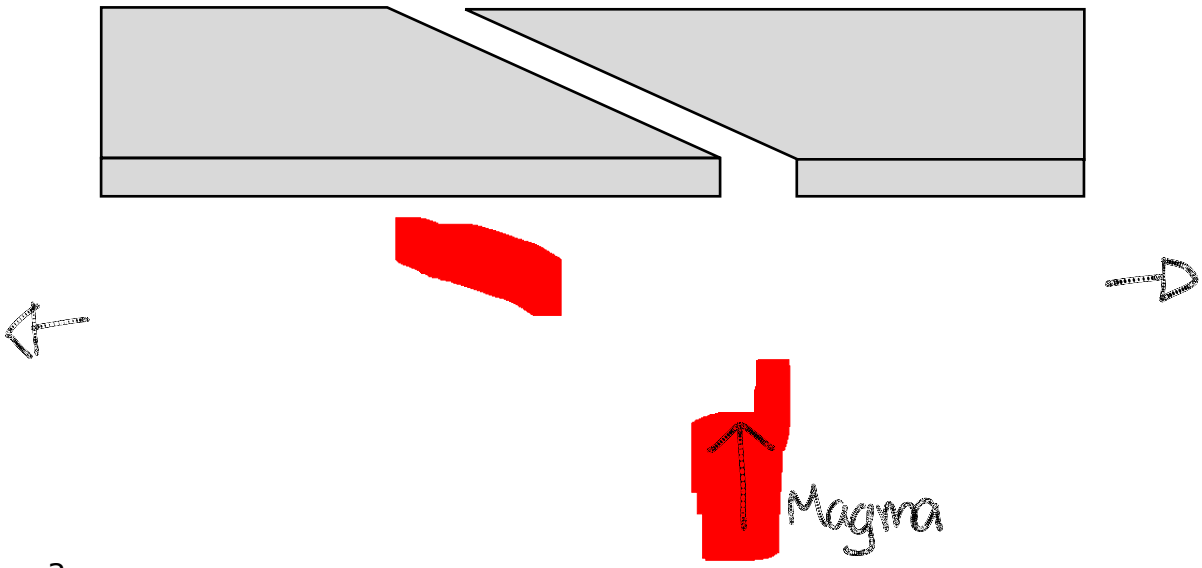
b. Divergent boundaries are also called constructive plate boundaries because this is where new crust is formed, from the cooling of magma

c. What are ocean ridges and how are they formed?

Wave-like rock that forms from the cooling of magma on the bottom of the ocean

d. Show or sketch **and label** on the diagram below:

- The direction of movement of divergent tectonic plates.
- Rising magma from the mantle
- Ocean ridges.



3.

a. Describe a convergent boundary.

Plates that move toward each other

b. Convergent boundaries are also called destructive plate boundaries because solid ocean crust melts into the mantle

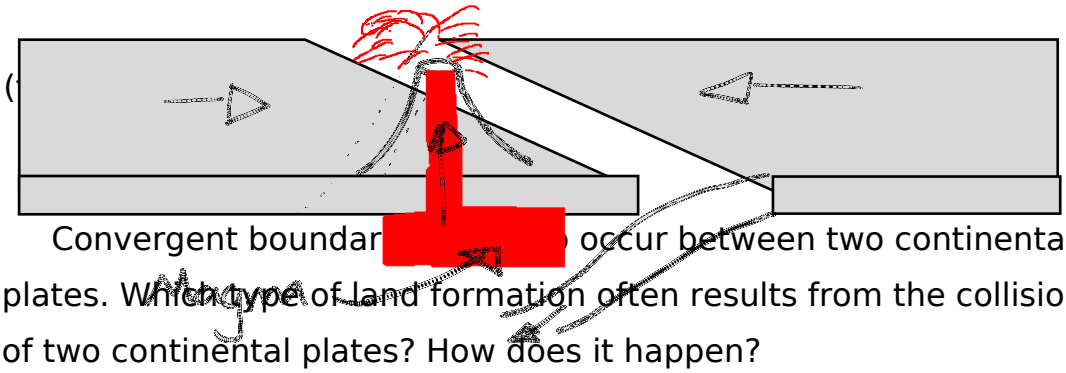
c. Subduction is when oceanic crust dives under the continental crust

d. How are volcanoes formed? The magma in the mantle bursts through the continental crust

e. Show or sketch **and label** on the diagram below:

- The direction of movement of convergent tectonic plates
- The oceanic plate subducting under the continental plate
- Solid crust becoming molten rock in the mantle
- The land formation often associated with convergent boundaries

4. (



Convergent boundar occur between two continental plates. Which type of land formation often results from the collision of two continental plates? How does it happen?

Mountain ranges. These form when two plates crumple upwards against each other

b. Name a well-known land formation that has resulted from this type of collision. Himalayas

5. (video and textbook)

a. Describe a transform boundary

Two neighbouring plates slide past each other. They also cause earthquakes

b. Transform boundaries are also called conservative plate boundaries because

Solid crust is neither created or destroyed

c. Think about the word ‘transform’. Why might the sliding segments of tectonic plates be called ‘transform boundaries’?

The landscape changes (transforms) as plates slide past

d. Do transform boundaries pose any dangers to humans and other creatures? Explain. Yes, they can cause earthquakes and tsunamis, which cause destruction of buildings, fires and potential loss of life.

e. Show or sketch **<u>and label</u>** on the diagram below:

- The direction of movement of transform plate boundaries
- Land formations that could be associated with transform boundaries

